Adversarial Self-Correction for Computational Research on Undeciphered Corpora: A Harness–Firewall–Refutation Architecture and Its Record

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Abstract

Computational study of undeciphered or otherwise ground-truth-free corpora is unusually prone to plausible-but-wrong conclusions: with no key against which to check a reading, "it comes out looking like language/a cipher/a number system" is nearly unfalsifiable, and the analyst's degrees of freedom are vast. We describe an operating architecture that couples three disciplines to blunt this: a synthetic *ground-truth harness* (every method must separate known positives from matched null/meaningless controls before it touches the real object); a *firewall* (every reported number is produced by deterministic, versioned code that records its own script, commit, inputs, and parameters—nothing is estimated or recalled); and a standing *clean-context adversarial refutation pass* in which an independent skeptic, working *without* the original analysis's context or investment, is tasked to build the strongest case against each finding, and no associational claim is graded reportable until it survives. We report the architecture's record over a ten-iteration computational program on the Voynich Manuscript (Beinecke MS 408). The pass caught a dozen-plus over-reads before they were reported—including conclusions asserted by the program's *own analysis code*—and, unusually, it overturned the program's own *negatives* as readily as its positives, once even refuting its own walk-back of a negative ("retraction, not reversal"). We argue that the transferable contribution is the architecture, not the case study, situate it against pre-registration, multiverse analysis, adversarial collaboration, and reproducible- research practice, and give an adoption checklist. The evidence is a single program's record—the main limitation—and, consistent with the method it describes, this paper invites the same refutation.

Keywords: adversarial validation; reproducible research; undeciphered scripts; red-teaming; researcher degrees of freedom; evidence grading; self-correction.

1 Introduction

The computational study of undeciphered corpora—the Voynich Manuscript, the Indus script, Linear A, rongorongo—has a recurring pathology: confident, mutually incompatible "solutions" are produced faster than they are refuted [Reddy and Knight, 2011, Bowern and Lindemann, 2021]. The root cause is structural. Where there is no ground truth, a result of the form "under transform T the text resembles class C" cannot be checked against a key; it can only be checked against *itself*, and a sufficiently flexible analysis will nearly always find some T that makes some C look plausible. This is the "garden of forking paths" [Gelman and Loken, 2013] and the researcher-degrees-of-freedom problem [Simmons et al., 2011] in an acute form: not merely many analyses of one dataset, but many analyses of an object whose correct analysis is unknown even in principle. The field's own literature supplies the cautionary archetype: the debate over whether block-entropy statistics show the Indus script to be linguistic [Rao et al., 2009] was settled methodologically by the demonstration that those very statistics *cannot* separate linguistic from non-linguistic symbol systems as used [Sproat, 2014]—a measure that does not discriminate the hypotheses it is invoked to decide.

Our response inverts the usual order of work. Rather than seek a finding and then defend it, we make the *validation machinery* the primary object and treat every candidate finding as guilty until an adversary fails to convict it. Concretely, we couple three disciplines—a harness, a firewall, and a standing clean-context refutation pass—so that the default output of the pipeline is not a claim but a *graded* claim that has already survived an attempt to destroy it, or an honest "not established." This paper describes that architecture (§2), explains why each component catches a distinct class of error (§3), and—its empirical core—reports the architecture's *record*: the catalogue of the program's own conclusions that the discipline overturned before they were published (§4). We then situate the approach against existing practice (§5), state its limitations reflexively (§6), and give an adoption checklist (§7). The case study is a ten-iteration program on the Voynich Manuscript; the numbers we quote from it are illustrative of *corrections*, and each traces to versioned code under the firewall. The claim of the paper, however, is not about the Voynich Manuscript. It is that this architecture is a transferable way to do defensible computational research where the answer cannot be looked up.

2 The architecture

Three disciplines, each necessary, jointly sufficient to make refutation bite.

2.1 The harness: validate on knowns before the unknown

No method is trusted on the real object until it behaves correctly on synthetic ground truth. For each method we assemble (i) *real positives* (period- and register-matched natural-language corpora), (ii) *matched negatives*—structured-but-meaningless generators (auto-copying/self-citation processes [Timm and Schinner, 2020], grille-style table generators, parameterised null models) tuned to the object's own nuisance parameters, and (iii) *cipher/transform controls*. A method that cannot separate these knowns is not used on the unknown, and a *replication gate* requires the pipeline to reproduce already-published baseline statistics before any novel experiment. The harness is what makes a negative result meaningful: "the statistic does not distinguish A from B" is only informative once the statistic is shown to distinguish a synthetic A from a synthetic B.

2.2 The firewall: numbers come only from code

Every statistic in every report is produced by deterministic, versioned code with fixed seeds, written to machine-readable output that records its producing script, commit hash, input-dataset version, and parameters. No number is estimated, remembered, or carried in prose. This is standard reproducible-research hygiene [Sandve et al., 2013], but in this architecture it is also load-bearing for *refutation*: because every claim dereferences to an exact computation, the adversary attacks the computation itself—re-running it, re-scoring it, probing its bands and nulls—rather than a summary sentence. A firewall turns "I don't find your argument convincing" into "your config `X` has its median out of band and you counted it as a match," which is checkable.

2.3 The standing clean-context refutation pass

Every finding is submitted to an independent skeptic instructed to build the *strongest* case against it, with positive and associational claims held to the harshest scrutiny (plausible-but-wrong positives being the field's dominant failure mode). The pass is *clean-context*: the refuter is given the claim, the code, and the data, but not the original analyst's narrative, hopes, or sunk cost. In our instantiation the refuter is a fresh-context large-language-model agent with read access to the repository and no memory of how the finding was reached; it returns a written brief—often several thousand words—quoting the offending numbers and recommending a grade. A claim is graded **A** (replicated/robust) or **B** (candidate) only after it survives; otherwise it is narrowed, downgraded to **C** (directional) or **D** (speculative), or withdrawn. This is the operational descendant of red-teaming and adversarial collaboration [Mellers et al., 2001], made cheap and repeatable enough to run on *every* finding rather than a chosen few.

2.4 Connective tissue: grades and a decision ledger

Two lighter rules bind the disciplines. *Evidence grading* (A–D) is attached to every claim, so downgrades are first-class outcomes rather than failures. *"Flag, don't resolve"*: any decision not covered by a locked ledger becomes a surfaced open item on the least-committal path, never a silent judgment call—so the forking paths are logged, not walked in secret. Together these convert the review of a program from "is the headline right?" to "is each claim carrying the grade its evidence and its refutation permit?"

3 Why it catches things

Each discipline structurally removes a distinct source of error.

The harness removes non-discriminating methods. A statistic that cannot separate the synthetic knowns is discarded before it can produce a spurious verdict on the unknown. This is exactly the failure the Indus entropy debate diagnosed [Rao et al., 2009, Sproat, 2014]; the harness makes it a precondition, not a post-hoc rebuttal.

The firewall removes recall drift and gives refutation teeth. Numbers that live only in prose accrete small errors and resist attack; numbers that dereference to code do neither. The firewall is what lets the refutation pass be adversarial in substance rather than rhetoric (§2.2).

The clean-context refuter removes confirmation bias and sunk cost. The analyst who built a finding is the worst-placed person to destroy it; an independent agent with no investment and no narrative is the best-placed. Running it on *every* claim—feasible only because the refuter is automated—turns adversarial review from an occasional event into a gate.

The consequence we most want to emphasise is *bidirectionality.* The replication-crisis literature concentrates on false positives [Ioannidis, 2005, Simmons et al., 2011], and for good reason. But in undeciphered-corpus work an over-strong *negative*—"no cipher of this class fits," "no generative family reproduces the signature"—is equally damaging and more insidious, because a negative silently closes a hypothesis and invites no replication. An architecture that only guards positives will let over-read negatives ship unchallenged. The record below shows this architecture overturning its own negatives as readily as its positives, and once refuting its own *correction* of a negative—which is, to our knowledge, an unusual degree of reflexive control.

4 The record

The empirical core of this paper is the catalogue of the program's own conclusions that the discipline overturned or narrowed before they were reported. Table 1 groups the salient cases by failure mode; the text then expands the most instructive. Every entry is a first-pass verdict that a less-disciplined pipeline would plausibly have published. Numbers trace to the program's versioned results under the firewall.

Circular positives and fitted points. The two most seductive errors were self-referential. In one iteration a joint-signature test reported that the manuscript was "favoured as a generation process"; the refuter observed that the only generator matching the manuscript was the one built to imitate it, so the match was true by construction and carried no evidence—the claim was narrowed to the robust *negative* it actually supported (a cipher-class exclusion). One iteration later the same trap reappeared in a new guise: a generator was graded a "class sufficiency" [**B**], until the refuter showed its constants had been grid-selected against the target's own statistical bands—a fitted point dressed as an a-priori draw—and that its weak-syntax criterion used a one-sided threshold that a fully order-shuffled control also passes. The corrected, band-honest test turned the apparent positive into a precise negative. Recognising the *same* circularity twice, across different experiments, is itself a product of the standing pass: the pattern was in the reviewer's checklist because it had been named once.

Mandatory nulls catch false positives that multiple-testing control does not. An anchor hunt for word→feature associations produced eighteen "hits." These survived Benjamini–Hochberg control yet were all false: the test was anti-conservative in a way FDR does not address, and a mandatory permutation null—shuffling the feature labels and re-running the entire pipeline—returned them to chance ($p \approx 0.48$). The lesson is that a null which re-executes the whole analysis is a different, stronger instrument than a p-value correction applied to its output.

Overturning negatives. Two cases show the bidirectional property. A candidate visual regularity was first over-claimed, then *over-killed* ("rater-idiosyncratic, KILLED"); the refuter showed the kill's central argument was a red herring and the honest status was *unresolved-underpowered*—the pipeline declining to over-read a negative exactly as it declines to over-read a positive. More strikingly, a whole iteration concluded that "no tested generative family reproduces the manuscript's joint signature." A refutation-directed follow-up showed the alleged obstruction was an artifact of a rigid morphology model; freeing the relevant control dissolved most of the coupling, and the negative was walked back. That walk-back was then *itself* refuted—no single configuration actually reproduced the full signature—yielding the final, doubly-corrected reading: "a retraction of the hard constraint, not a promotion to no constraint." The pipeline over-read a negative, corrected it, and corrected the correction.

Corrections can also strengthen. Not every refutation downgrades. A cipher-class exclusion was first established over a single plaintext language; the refuter demanded typologically diverse controls; adding them *upgraded* the exclusion from language-specific to universal. Adversarial review improves calibration in both directions.

Across ten iterations the discipline produced, in addition to the graded findings, a running log of these overturns—conclusions asserted by the analysis, and in several cases by the analysis *code*, that were withdrawn or narrowed before they reached a reader. We regard that log, not any individual manuscript finding, as the program's most robust output.

5 Relation to existing practice

The components are individually familiar; the contribution is their integration and the gate they jointly enforce. *Pre-registration* [Nosek et al., 2018] fixes analyses in advance to curb forking paths, but cannot help an exploratory program on an object whose right analysis is unknown; our decision ledger and standing refutation are the exploratory analogue—forking paths are logged and adversarially reviewed rather than pre-committed. *Multiverse* [Steegen et al., 2016] and *specification-curve* [Simonsohn et al., 2020] analyses report a claim across many defensible specifications; our harness-gated, band-honest scoring is a targeted version—and the refuter's job is precisely to find the specification that breaks the headline. *Adversarial collaboration* [Mellers et al., 2001] pairs opposed experts; the clean-context automated refuter is a scalable, sunk-cost-free instantiation that can be run on every claim rather than a negotiated few. *Reproducible-research* practice [Sandve et al., 2013] supplies the firewall; we add that reproducibility is not only an end (others can re-run) but a *means* (the adversary can attack the computation). The novel combination is thus: harness-gated methods, a firewall that gives refutation teeth, a standing clean-context refuter as a publication gate, and—most distinctively—bidirectional and recursive self-correction, including of the program's own negatives and of its own corrections. The automated fresh-context refuter is the enabling ingredient: it makes adversarial review cheap enough to be a default rather than an event.

6 Limitations and reflexivity

The evidence is a *single program's record*. We cannot present a controlled counterfactual—the same program run without the discipline—so we cannot quantify how many of the overturned claims would truly have been published elsewhere; we can only show that they were the pipeline's genuine first-pass verdicts. The architecture reduces but does not eliminate error: the refuter can miss things, grades are judgment calls, and there is a real risk of *over*-refutation—killing true findings—for which the double-corrected root↔leaf case is both a cautionary instance and a demonstration that the tension is itself detectable. The approach is not free: each refutation is a full independent analysis, and the firewall imposes an engineering tax. The automated refuter inherits its substrate's blind spots; an adversary and analyst drawn from the same model family may share failure modes, so cross-vendor or human refutation remains the stronger check where stakes are high. Finally, and in keeping with the method it advocates, this paper is written under the firewall it describes and makes claims that should themselves be refuted; we regard an adversarial pass on *this* argument as the appropriate next step, not an optional one.

7 Adoption checklist

For a computational program on an ambiguous or ground-truth-free corpus:

1. **Build the harness first.** Real positives, matched structured-meaningless negatives, and transform/cipher controls. Do not run a method on the object until it separates the knowns; gate novel work behind reproducing published baselines.
2. **Erect the firewall.** Every reported number from deterministic, versioned code that stamps its script, commit, inputs, and parameters. No number in prose that is not in a results file.
3. **Grade everything A–D**, and make downgrades and "not established" first-class outcomes.
4. **Run a clean-context refutation pass on every A/B claim**—an independent reviewer, without the analyst's context, tasked to break it, with authority to downgrade. Hold negatives

to the same standard as positives.

5. **Log decisions; don't resolve silently.** Surface every uncovered choice on the least-committal path.
6. **Refute your corrections.** A walk-back is a claim; subject it to the same pass.
7. **Diversify the adversary** (cross-vendor model, or human) where stakes justify it.

8 Conclusion

Where the answer cannot be looked up, the discipline of *how* a claim is made must carry the weight that ground truth carries elsewhere. Coupling a validating harness, a firewall that makes every number attackable, and a standing clean-context adversarial pass produces a pipeline whose default output is a graded claim that has already survived an attempt to destroy it—and, as the record shows, a pipeline willing to overturn its own positives, its own negatives, and its own corrections. The transferable result is not a reading of any manuscript; it is that adversarial self-correction, made cheap enough to run on every claim, is a practical route to defensible computational research on undeciphered corpora.

Reproducibility

The case-study program's statistics are produced by deterministic, versioned code (fixed seeds) writing machine-readable results, each recording its producing script, commit, dataset version, and parameters; the self-correction record is documented in the program's synthesis notes. Numbers quoted here are illustrative of corrections and trace to that code.

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Failure mode	**First-pass claim (would have shipped)**	**What caught it**	**Direction**
Circular positive	"The manuscript is favoured as a *generation* process" (E19)	Refuter: the only matching generator was tuned to the target; match is by construction	false positive
Fitted-to-target	"The generative class is *sufficient*" [B] (E21)	Refuter: constants were grid-selected to hit the target bands (a fitted point mislabelled a-priori); weak-syntax test used a threshold a shuffle also passes	false positive
Missing null / control	18 word→feature "anchors" (E7); "needs constructed morphology" (E6); "whitening confirms" (E8)	Mandatory permutation null (E7); the abjad control the refuter demanded (E6); covariance shown ill-conditioned (E8)	false positive
Broken metric / artifact	A "meaning certificate" from a word-order statistic; a trivial-endpoint effect (E9)	Two axes shown broken; a pre-registered commitment retracted	false positive
Confound	Currier A/B "differ in grammar" (E14)	Content-controlled study: the difference reverses within-section (E17)	false positive
Over-strong negative	"root↔leaf bundle KILLED" (E12); "no generative family reproduces the signature" (i08)	Refuter: the kill's key argument was a red herring → "unresolved" (E12); the coupling was a morphology artifact, walked back (i09)	**false negative**
Recursive (self-check)	The i09 *walk-back* itself: "reproduces all axes / signature doesn't constrain mechanism"	Refuter of the walk-back: no single config matches all axes → "retraction, not reversal"	over-correction
Demanded-control upgrade	A Latin-only exclusion (E19)	Refuter demanded typologically diverse controls; exclusion *upgraded* to universal (E19b)	strengthening

Table 1: Selected entries from the program's self-correction record (iterations i03–i09). Each first-pass claim was overturned, narrowed, or strengthened by the discipline before publication. Note the recurrence of the *false-negative* and *recursive* rows— corrections in the direction the replication-crisis toolkit usually ignores.